PYTHON MENU-DRIVEN PROJECTS

List, Set, Tuple and Dictionary

Four simple, copy-paste runnable Python projects for beginners. Each project uses a loop, menu options, conditions and collection operations.

Collection	Project	Filename
List	Shopping List Management	list_project.py
Set	Unique Course Management	set_project.py
Tuple	Fixed Product Information	tuple_project.py
Dictionary	Student Management System	dictionary_project.py

How to run: Save each program using its given filename. Open the terminal in VS Code and run `python filename.py`.

1. List Menu-Driven Project

Project: Shopping List Management
Filename: `list_project.py`

This project stores shopping items in a list. The user can add, view, update, remove and search items.

Complete Program

```
shopping_list = []

while True:
    print("\n===== SHOPPING LIST MENU =====")
    print("1. Add Item")
    print("2. View Items")
    print("3. Update Item")
    print("4. Remove Item")
    print("5. Search Item")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        item = input("Enter item name: ")
        shopping_list.append(item)
        print(item, "added successfully.")

    elif choice == "2":
        if len(shopping_list) == 0:
            print("Shopping list is empty.")
        else:
            print("\nShopping Items:")
            for index, item in enumerate(shopping_list, start=1):
                print(index, ".", item)

    elif choice == "3":
        if len(shopping_list) == 0:
            print("Shopping list is empty.")
        else:
            for index, item in enumerate(shopping_list, start=1):
                print(index, ".", item)

            number = int(input("Enter item number to update: "))

            if 1 <= number <= len(shopping_list):
                new_item = input("Enter new item name: ")
                shopping_list[number - 1] = new_item
                print("Item updated successfully.")
            else:
                print("Invalid item number.")

    elif choice == "4":
        item = input("Enter item name to remove: ")

        if item in shopping_list:
```

```
            shopping_list.remove(item)
            print(item, "removed successfully.")
        else:
            print("Item not found.")

    elif choice == "5":
        item = input("Enter item name to search: ")

        if item in shopping_list:
            print(item, "is available in the list.")
        else:
            print(item, "is not available.")

    elif choice == "6":
        print("Thank you!")
        break

    else:
        print("Invalid choice. Please try again.")
```

Methods and Concepts Used

Method / concept	Use
append()	Adds a new shopping item
remove()	Removes an item using its value
enumerate()	Displays items with serial numbers
in	Checks whether an item exists
list[index]	Updates an item using its position

Important: Copy the complete code exactly, save the file, and run it from the VS Code terminal.

2. Set Menu-Driven Project

Project: Unique Course Management
Filename: `set_project.py`

This project stores unique course names in a set. Duplicate course names are not allowed.

Complete Program

```
courses = set()

while True:
    print("\n===== COURSE MANAGEMENT MENU =====")
    print("1. Add Course")
    print("2. View Courses")
    print("3. Remove Course")
    print("4. Search Course")
    print("5. Count Courses")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        course = input("Enter course name: ")

        if course in courses:
            print("Course already exists.")
        else:
            courses.add(course)
            print(course, "added successfully.")

    elif choice == "2":
        if len(courses) == 0:
            print("No courses available.")
        else:
            print("\nAvailable Courses:")
            for course in courses:
                print("-", course)

    elif choice == "3":
        course = input("Enter course name to remove: ")

        if course in courses:
            courses.remove(course)
            print(course, "removed successfully.")
        else:
            print("Course not found.")

    elif choice == "4":
        course = input("Enter course name to search: ")

        if course in courses:
            print(course, "is available.")
        else:
            print(course, "is not available.")
```

```
    elif choice == "5":
        print("Total unique courses:", len(courses))

    elif choice == "6":
        print("Thank you!")
        break

    else:
        print("Invalid choice. Please try again.")
```

Methods and Concepts Used

Method / concept	Use
set()	Creates an empty set
add()	Adds one unique course
remove()	Removes an existing course
len()	Counts unique courses
in	Checks whether a course exists

Important: Copy the complete code exactly, save the file, and run it from the VS Code terminal.

3. Tuple Menu-Driven Project

Project: Fixed Product Information
Filename: `tuple_project.py`

This project stores fixed product information in a tuple. The values can be viewed and searched but cannot be modified.

Complete Program

```
product = (
    101,
    "Laptop",
    55000,
    "Electronics",
    "Available"
)

while True:
    print("\n===== PRODUCT MENU =====")
    print("1. View Complete Product")
    print("2. View Product ID")
    print("3. View Product Name")
    print("4. View Product Price")
    print("5. Search Value")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        print("\nProduct Details")
        print("Product ID:", product[0])
        print("Product Name:", product[1])
        print("Price:", product[2])
        print("Category:", product[3])
        print("Status:", product[4])

    elif choice == "2":
        print("Product ID:", product[0])

    elif choice == "3":
        print("Product Name:", product[1])

    elif choice == "4":
        print("Product Price:", product[2])

    elif choice == "5":
        value = input("Enter a value to search: ")
        found = False

        for item in product:
            if value.lower() == str(item).lower():
                found = True
                break
```

```
        if found:
            print(value, "is available in the tuple.")
        else:
            print(value, "is not available.")

    elif choice == "6":
        print("Thank you!")
        break

    else:
        print("Invalid choice. Please try again.")
```

Methods and Concepts Used

Method / concept	Use
product[index]	Reads a value using its position
for loop	Checks every value in the tuple
str()	Converts a value to text for searching
lower()	Performs case-insensitive comparison
Immutable	Tuple values cannot be changed

Important: Copy the complete code exactly, save the file, and run it from the VS Code terminal.

4. Dictionary Menu-Driven Project

Project: Student Management System
Filename: `dictionary_project.py`

This project stores student records using roll numbers as keys. The user can add, view, search, update and remove students.

Complete Program

```
students = {}

while True:
    print("\n===== STUDENT MANAGEMENT MENU =====")
    print("1. Add Student")
    print("2. View All Students")
    print("3. Search Student")
    print("4. Update Student")
    print("5. Remove Student")
    print("6. Exit")

    choice = input("Enter your choice: ")

    if choice == "1":
        roll_number = input("Enter roll number: ")

        if roll_number in students:
            print("Student already exists.")
        else:
            name = input("Enter student name: ")
            course = input("Enter course name: ")
            marks = float(input("Enter marks: "))

            students[roll_number] = {
                "name": name,
                "course": course,
                "marks": marks
            }
            print("Student added successfully.")

    elif choice == "2":
        if len(students) == 0:
            print("No student records available.")
        else:
            print("\nStudent Records:")

            for roll_number, details in students.items():
                print("\nRoll Number:", roll_number)
                print("Name:", details["name"])
                print("Course:", details["course"])
                print("Marks:", details["marks"])

    elif choice == "3":
        roll_number = input("Enter roll number to search: ")
        student = students.get(roll_number)
```

```
        if student:
            print("Name:", student["name"])
            print("Course:", student["course"])
            print("Marks:", student["marks"])
        else:
            print("Student not found.")

    elif choice == "4":
        roll_number = input("Enter roll number to update: ")

        if roll_number in students:
            students[roll_number]["name"] = input("Enter new name: ")
            students[roll_number]["course"] = input("Enter new course: ")
            students[roll_number]["marks"] = float(input("Enter new marks: "))
            print("Student updated successfully.")
        else:
            print("Student not found.")

    elif choice == "5":
        roll_number = input("Enter roll number to remove: ")

        if roll_number in students:
            students.pop(roll_number)
            print("Student removed successfully.")
        else:
            print("Student not found.")

    elif choice == "6":
        print("Thank you!")
        break

    else:
        print("Invalid choice. Please try again.")
```

Methods and Concepts Used

Method / concept	Use
dictionary[key]	Adds or updates a record
items()	Reads all keys and values
get()	Searches safely using a key
pop()	Removes a record
in	Checks whether a roll number exists

Important: Copy the complete code exactly, save the file, and run it from the VS Code terminal.

Project Summary

Collection	Project	Main Operations
List	Shopping List	Add, view, update, remove, search
Set	Course Management	Add unique, view, remove, search, count
Tuple	Product Information	View fixed values and search
Dictionary	Student Management	Add, view, search, update, remove

Common Program Flow

1. Create the collection.
2. Display the menu inside a `while True` loop.
3. Read the user's choice.
4. Use `if-elif-else` to perform an operation.
5. Use `break` when the user selects Exit.

Suggested Practice

Run every project once. Then change the project data: products instead of shopping items, employee skills instead of courses, a book instead of a product tuple, or employee records instead of student records.